Projects, Papers, and Parts

Do student iGEM projects predict local academic research in synthetic biology?

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Why this is interesting

- iGEM is a global natural experiment in distributed innovation
- 100k+ participants; 400+ universities; 50+ countries
- Projects, parts, and wiki pages are publicly archived
- Runs annually – a repeated cross-section from 2009 to 2025

The iGEM corpus presents us with unique fine-grained insights into a normally hidden process of innovation

Are student projects topically embedded in local research?

Inspired by:

- Hidalgo et al. (2007): revealed comparative advantage and product space
- Neffke & Henning (2013): skill-relatedness from labor flows

Applied here to text, not products or occupations.

- Unit: city
- Similarity: cosine distance in embedding space
- Question: do student projects and local papers occupy related regions of semantic space?

Core hypothesis

Innovation follows a local path: student projects $\rightarrow$ papers $\rightarrow$ patents

If so: project topics should *lead* paper topics in the same city by a few years.

Data

Corpus	Docs	Cities	Years	Source
Synbio papers	10,402	1,218	2000–2026	OpenAlex + PMC
iGEM projects	4,614	626	2009–2025	iGEM Teams
BioBrick parts	89,060	472	2009–2025	iGEM Parts Registry
Analysis sample	–	**387**	–	cities with both

Carbon-capture tagged: 294 papers (2.8%) and 141 projects (3.1%).

SPECTER2 embeds scientific text into a citation-informed space

- 768-dim vectors; contrastive learning on citation pairs
- **Proximity adapter**: fine-tuned for semantic similarity, not classification
- Cached for all 15,000 documents; fully reproducible

City representation:

$$\bar{\mathbf{a}}_c = \frac{1}{|P_c|} \sum_{p \in P_c} \mathbf{e}_p$$

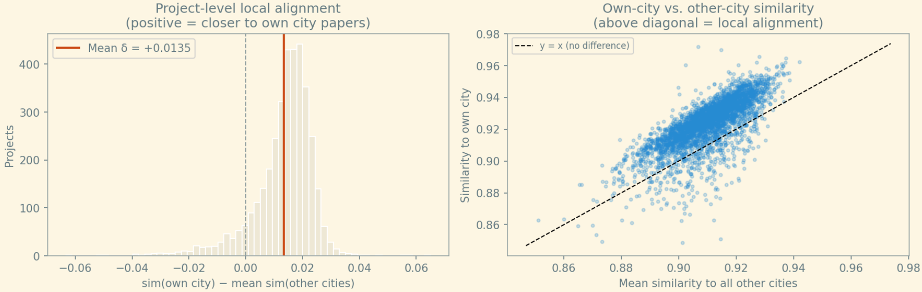
Mean embedding over all docs of type t in city c.

Note: mean embeddings assume a single topical centroid per city. Topic-cluster analysis (later) provides a complement that does not require this assumption.

Why SPECTER2?

Standard sentence transformers are not calibrated for scientific text. SPECTER2 is trained on 146M citation pairs; documents that co-cite tend to be nearby.

90% of projects are semantically closer to their own city's papers



n = 3,620 projects; cities with $\geq$ 3 papers; δ = own-city cosine − mean over all other cities

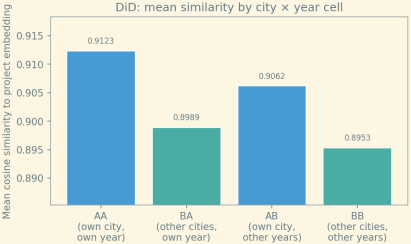
Local Alignment Statistical Tests

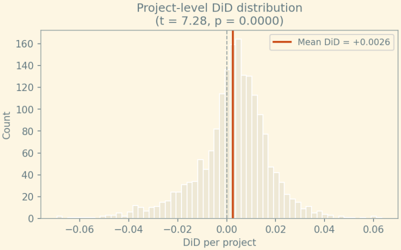
	Statistic	p-value
Mean δ (own − other)	$+0.0135$	
Fraction with $\delta > 0$	90.6%	
One-sample t-test	$t = 78.02$	< 0.001
Wilcoxon signed-rank		< 0.001

Takeaway

Projects are systematically closer to local papers than to papers anywhere else. The effect is extremely consistent.

DiD: separating stable specialisation from temporal alignment





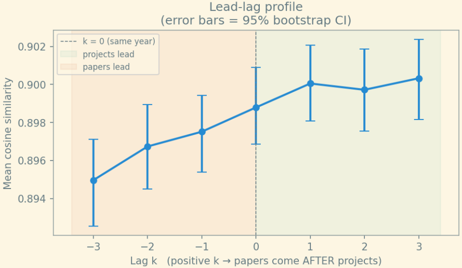
Annual city-year centroids ($\geq$ 2 papers); 1,841 projects in DiD sample

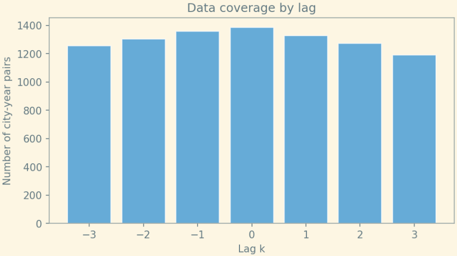
The temporal component is real

	Own city	Other cities	Local advantage
Own year	0.9123	0.8989	$+0.0135$
Other years	0.9062	0.8953	$+0.0109$
Δ (temporal)	$+0.0062$	$+0.0036$	
DiD		$\mathbf{+0.0026}$	

Stable city specialisation accounts for most of the alignment. The DiD component is small but positive.

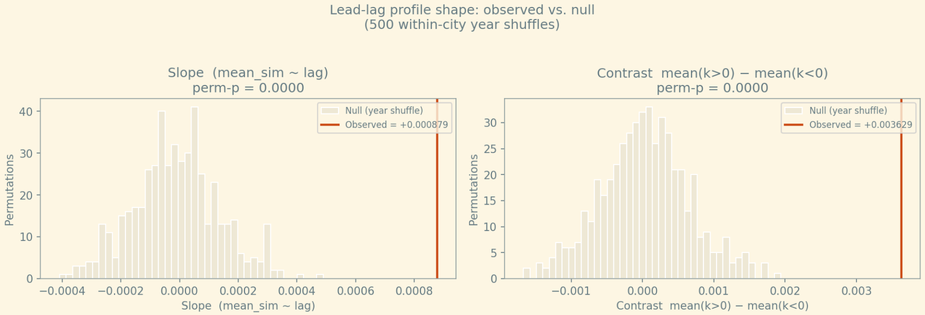
Projects are semantically ahead of local papers





Lag k: cosine similarity between project centroid in year t and paper centroid in year $t+k$; bootstrapped 95% CI

The lead-lag slope is not a temporal sampling artifact



500 within-city year-shuffles; observed slope $= +0.000879$; null $p_{95} = +0.000246$; perm-$p < 0.05$

Topic clusters: mapping synthetic biology subfields

- UMAP reduces 768-dim embeddings to 60 dimensions (cosine metric)
- HDBSCAN finds **83 clusters** across all 15,000 documents
- ∼29% of documents are labelled "noise" — HDBSCAN deliberately leaves ambiguous points unassigned rather than forcing them into a cluster
- Cluster overlap = weighted Jaccard on cluster frequency vectors
- Sample: 189 cities with ≥ 3 clustered docs per type

Mean cluster overlap: 0.081; median: 0.023; max: 0.717

Cluster co-membership analysis

1. For each city, compute a **cluster frequency distribution** — the share of that city's papers (and separately, projects) in each cluster.
2. Define **cluster co-membership** as the cosine similarity between those two distributions.
3. Test whether co-membership is locally specific under within-country city permutation.

Project-level test: projects enter clusters already known to their city

For each non-noise project in cluster k, city c, year y:

$\texttt{in\_known} = \mathbf{1}[\text{city } c \text{ has a paper in cluster } k \text{ before year } y]$

	Value
Projects in analysis	3,004
Own-city known rate	3.1%
Mean baseline (random city)	0.6%
Mean δ (own − baseline)	$+0.025$
t-test (H_0: $\delta = 0$)	$t = 7.99$
p-value	< 0.001

Takeaway

Projects are $\approx 5\times$ more likely to enter a cluster their city already publishes in than a randomly assigned city would be. The signal survives because the test is project-level with a temporal direction, not an aggregate city-level score.

iGEM gives us two independent windows on what a city's teams care about

Lens 1 — text (wiki pages):

- What did teams write about?
- Captured by SPECTER2 embeddings and topic clusters

Lens 2 — physical parts (BioBricks):

- What DNA components did teams actually build and register?
- Types: promoter, CDS, RBS, terminator, reporter, device, ...
- City part-type profile: proportion vector over part types

These are *independent* measurement systems. If they agree, that is evidence of a coherent underlying signal.

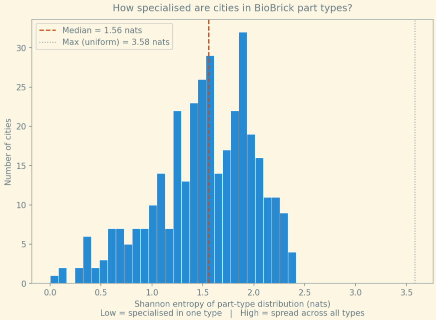
Mantel test (179 cities, 999 permutations)

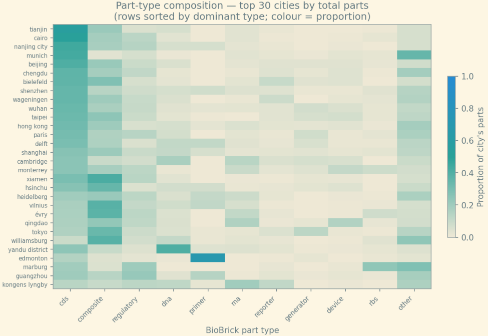
$$r = 0.2207, \quad p < 0.001$$

Null $p_{95} = 0.073$.

Part-type distance and semantic cluster distance are correlated. Teams that write about similar things also build similar biological tools.

Part-type specialisation varies meaningfully across cities





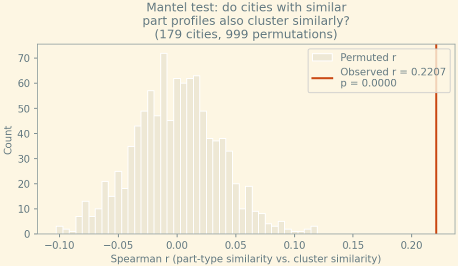
Top 30 cities by total registered parts; sorted by dominant part type

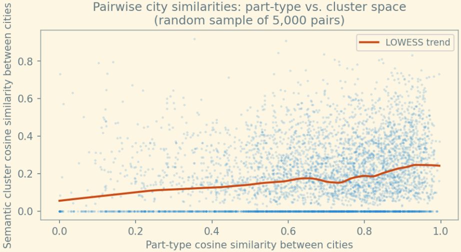
Mantel test: comparing two pairwise similarity matrices

Standard method from ecology for asking whether two similarity structures agree.

1. For every pair of cities (i, j), compute:
   - `sim_parts`(i, j): cosine similarity between their part-type vectors
   - `sim_cluster`(i, j): cosine similarity between their cluster-frequency vectors
2. Collect all $N \times (N-1)/2$ such pairs (upper triangle of each matrix).
3. Compute the **Spearman rank correlation** between the two sets of similarities. A positive correlation means: cities similar in part-type space are also similar in cluster space.
4. **Permutation null**: shuffle the rows and columns of one matrix randomly many times. This destroys the city labels while preserving the matrix structure. The p-value is the fraction of shuffles that produce a correlation as high as observed.

Mantel test: part-type and cluster distances are correlated





Spearman Mantel $r = 0.2207$; perm-$p < 0.001$; 179 cities

Results

1. **Local alignment is real and consistent** – 90.6% of projects are topically closer to their own city's papers ($t = 78$, $p < 0.001$).
2. **Projects lead papers by ≈ 3 years** – lead-lag peak at $k = +3$; profile shape non-random under permutation.
3. **Projects enter clusters their city already knows** – own-city cluster entry rate $5\times$ baseline; $t = 7.99$, $p < 0.001$; holds with temporal direction.
4. **Part-type space corroborates semantic space** – Mantel $r = 0.22$, $p < 0.001$; two independent signals agree.

Limitations and what comes next

Current limitations:

- No causal identification – shared local environment could drive both
- Temporal relationships difficult to estimate due to unclear research timelines
- iGEM Projects and papers use different language, SPECTER2 not trained on project data
- Patents not yet integrated

Next steps:

- Integrate patent corpus
- Research success; connect to citation count for papers and awards for projects
- Fine-tune model based on links from biobrick registry (paper-part-project)
- Improve interactive semantic space explorer + geographic view

Takeaway

- Student iGEM projects are semantically correlated with local research, and projects are **approx 5 times** more likely to enter a topic cluster their city already publishes in.
- Text embeddings and BioBrick composition point to the same city-level topical structure.
- Projects share maximum similarity with local papers roughly 3 years later. Causal interpretation remains open.